Overview of Three Projects

Disaster Impact Classification



- Objective: Classify disasters into low-impact and high-impact categories.
- Dataset & Preprocessing: EM-DAT Dataset · Disaster characteristics and human impacts.
- Models: Artificial Neural Network (ANN) · Logistic Regression baseline.
- Evaluation & Results: Logistic Regression: Accuracy 0.77 · ANN: Recall 0.33, F1-score 0.40 for the high-impact class.
- Potential Applications in Urban Resilience: Disaster Impact Assessment

Yolo Urban Object Detection



- Objective: Detect common objects in urban images
- Input Image: Selected urban images (City. jpg).
- Methodology: Pretrained YOLO · Inference only · No fine-tuning
- Results & Limitations: Object classes, bounding boxes, confidence scores.
- Potential Applications in Urban Resilience: Urban Damage & Infrastructure Monitoring

House Price Prediction



- Objective: Predict house prices and compare regression models.
- Dataset & Preprocessing: Seattle Housing Dataset · Data cleaning and preprocessing
- Models: Linear Regression · Random Forest · XGBoost
- Evaluation & Results: MAE · MSE · R^2
- Potential Applications in Urban Resilience: Housing Vulnerability & Recovery Cost Analysis